Therapeutic Innovation & Regulatory Science (2024) 58:845–854
https://doi.org/10.1007/s43441-024-00639-0

ORIGINAL RESEARCH

Simultaneous Global Drug Development and Multiregional Clinical Trials (MRCT): 5 Years After Implementation of ICH E17 Guidelines

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Received: 23 October 2023 / Accepted: 1 March 2024 / Published online: 12 May 2024

Abstract

The ICH E17 guidelines (2014–2017) on Multiregional Clinical Trials (MRCT) was a joint effort by the regulators and industry to facilitate simultaneous global drug development and registration through taking a strategic approach for clinical trials. In other words, the objective was to reduce the time it takes to bringing medications to patients around the world through minimizing unnecessary duplication of local or regional studies, which may add the regulatory burden to cost and time of bringing new therapies to patients. Under the auspices of ICH, training materials were created and provided to various stakeholders. Despite the successful promotion of the benefits of ICH E17 MRCT guidelines across the different regions, the uptake of some concepts (e.g., pooling strategy) in the ICH E17 guidelines has been slow. This paper describes various factors which could affect the conduct of MRCT at a global level, including ambiguity in definition of "region" (in MRCT), new regulatory requirements to enroll a diverse patient population, the use of decentralized clinical trials, use of data sources other than randomized clinical trials (e.g., use of Real Word Data), and the impact of the COVID-19 pandemic on the conduct of MRCT.

Keywords Multiregional clinical trials · MRCT · Drug development · ICH E17

Introduction

History of ICH E17 Guidelines

The International Council for Harmonization (ICH) is a global organization that brings together regulatory authorities and pharmaceutical industry experts from around the world to develop guidelines on the quality, safety, and efficacy of pharmaceuticals. One of the recent guidelines developed by the ICH is the E17 guideline, which focuses on the general principles for planning and design of multi-regional clinical trials (MRCTs) in drug development.

The development of the ICH E17 guideline began in 2014 with the formation of an expert working group consisting of regulatory authorities and industry experts from across the world. The working group was tasked with developing guidelines to harmonize the planning and design of MRCTs simultaneously across multiple regions. After several years of discussions and revisions, the draft guideline was endorsed by the ICH assembly in June 2016 followed by the public consultation in each region. The guideline was revised based on feedback received from stakeholders, and the final version was published in November 2017 [1].

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The guideline aims to ensure that MRCTs are designed and conducted in a manner that produces reliable and interpretable results while minimizing the burden on study subjects and investigators. Subsequently, the ICH sponsored the E17 Implementation Working Group to develop training material and provide the training to various stakeholders including the regulators, academia, and the industry [2].

The ICH E17 guideline, expanding on the ICH E5 [3], covers several key topics, including the rationale for conducting a MRCT, the selection of regions and populations for inclusion, the use of bridging studies, and the handling of data from different regions. It is recommended that the ICH E17 should be used together with other ICH guidelines, including E5, E6, E8, E9, E10 and E18 guidelines [2]. By providing clear and comprehensive guidance on the planning and design of MRCTs, the guidelines make sure that clinical trial data is reliable and interpretable, ultimately benefiting patients and public health.

Status of Implementation of ICH E17 Guidelines in each Region

A literature search on PubMed for the term 'MRCT' or multiregional clinical trials show that the number of references peaked around 2020–2021. The US Food and Drug Administration adopted the guidance in July 2018. The European Federation of Pharmaceutical Industries and Associations (EFPIA) found in its survey of seven multinational companies that there was significant difference in uptake and application of ICH E17 guideline [4]. A similar survey by the Japanese Pharmaceutical Manufacturing Association (JPMA) found that the respondents had questions about the application of the MRCT guidelines [5]. In both the EU and Japan surveys, lack of clarity on defining the sample size and "pooling strategies" was identified as one of the major reasons on lack of understanding of the guidelines.

A publication of an online survey of representatives from industry, academic centers and clinical research organizations in Korea [6] evaluated the ICH E17 guideline for clarity, applicability to Korea, the impact to local environment and the adoption time. The acceptability and appropriateness of the guidelines varied across various groups of respondents. The results suggest adoption period was approximately 1–2 years.

In Japan [7], dose-selection, consistency between the overall- and Japanese-populations and good clinical practice compliance were well considered in MRCTs. However, the use of relevant guidelines for disease and primary end point definitions, standardization of efficacy/safety information, sample size allocation as well as training/validation on subject selection and primary end point, were addressed less adequately and may need to be considered when planning future MRCTs. Recent studies also suggest further collaboration opportunity in Asia in conducting MRCTs [8] and a potential delay in drug approvals in Japan resulting from less participations of Japan in MRCTs [9].

These findings suggest that there may be a need to improve awareness and adherence to the ICH E17 guideline at a global level to ensure that MRCTs are conducted efficiently and with high-quality data that can be used for drug approval across multiple regions. As depicted in Fig. 1 from the ICH E17 guidelines, design and conduct MRCTs will allow for near simultaneous registration of drugs across multiple regions [2].

Possible Reasons Behind Slow Uptake of ICH E17 Guidelines

Over the last decade, particularly since the outbreak of COVID-19 pandemic, traditional drug development paradigm is changing. Historically, drug development applied the sequential Ph1-2-3 drug development paradigm. This traditional approach is changing due to several factors that have emerged over the past few years. This is particularly true for diseases with significant unmet medical needs, such as rare diseases, life-threatening conditions, or areas where existing therapies are inadequate or non-existent. These factors have led to a shift in the way new drugs are discovered, developed, and approved. The ICH E17 guidelines as written may not take into account changes in the drug development paradigm which are discussed below.

Increased Focus on Personalized Medicine

The growing understanding of the human genome and the molecular basis of diseases has paved the way for personalized medicine. This approach aims to develop targeted therapies based on a patient's unique genetic makeup, which may render the traditional one-size-fits-all approach obsolete.

Expedited Regulatory Pathways

Expedited regulatory pathways including the US FDA's fast track, breakthrough therapy, priority review designations, and accelerated approvals [10], EU's PRIME designation [11] or PMDA's priority review and Sakigake designation and conditional approvals [12] are mechanisms provided by regulatory agencies to facilitate and expedite development and review of new drugs that are intended to address an unmet medical need. These pathways aim to ensure that therapies for serious conditions are approved and available

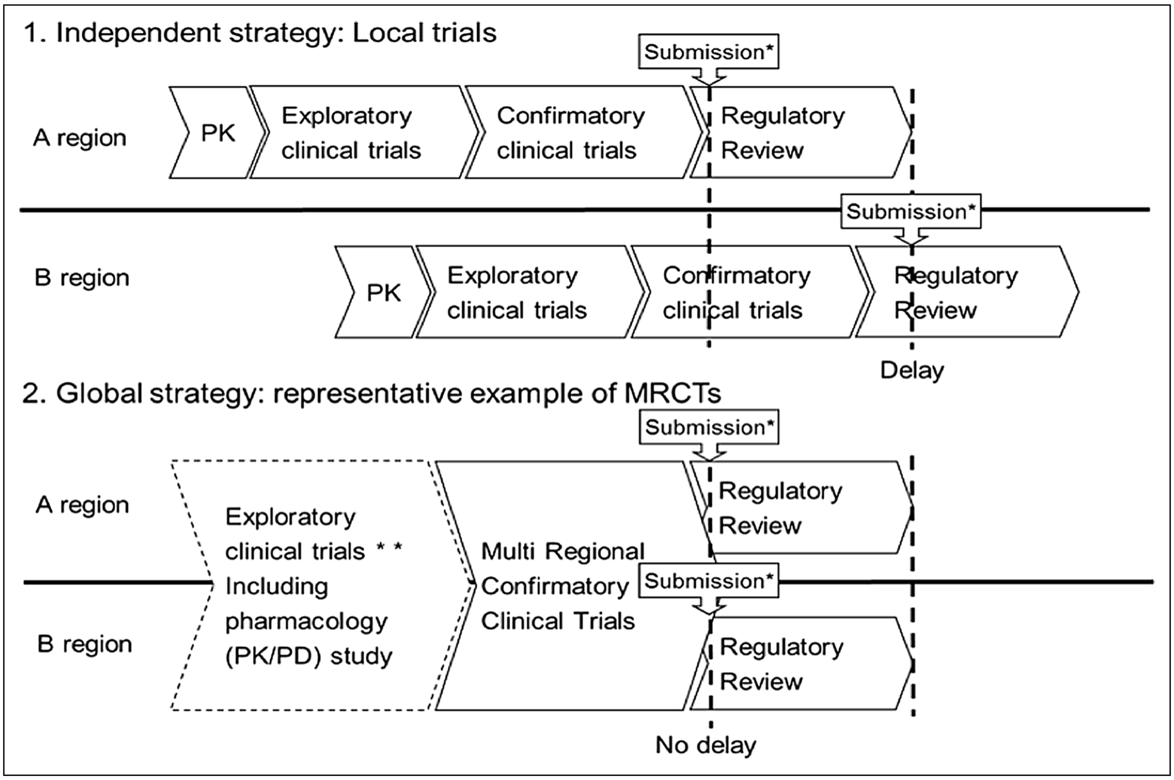

Figure 1 Conduct of MRCTs could lead to near simultaneous regulatory submissions (adapted from the ICH E17 guideline [1]).

to patients as soon as possible. To take advantage of these expedited pathways, the sponsors tend to conduct local or regional studies rather than conducting MRCTs as these mechanisms allow for faster regulatory approval based on Phase 1 or Phase 2 data, which can be especially critical for patients with limited or no treatment options which are not generally amenable to MRCT as described in the E17 guideline.

Focus on Patient-centricity

There is a growing emphasis to involve patients in the drug development process, from early-stage research to post-approval studies. This shift towards patient-centricity ensures that new therapies are developed based on patients' needs and preferences. While the use of patient reported outcomes (PROs) in MRCTs can provide valuable insights into the effectiveness and impact of treatments from the patient's perspective, there are several challenges associated with their use such as cultural and linguistic differences which can lead to challenges in evaluating results of PRO measures and assuring the reliability and validity of the data. Translation of PRO measures needs to be culturally appropriate, not just linguistically correct. Depending on the setting and the patient population, different methods of data collection (e.g., paper, electronic, interview) may be utilized. The collection methods may have implications for data quality, sensitivity and specificity of PRO measurements. Finally, ensuring the same PRO measures are analyzed and interpreted in the same way across diverse regions and populations can be challenging, making it difficult to interpret the MRCT data generated from PROs. Although there is no specific mention of PROs in the current ICH E17 guidelines which is an evolving area in conduct of clinical trials, the E17 did discuss the use of endpoints that may have differences in understanding when used in different cultural settings.

Change in Definition of a "Region" and "Pooled Region"

In the context of MRCTs [2], "Region" is typically defined based on a combination of factors such as:

Geography

This is the most obvious factor and refers to the physical location of the countries involved. For instance, the region might be "East Asian countries" or "Sub-Saharan African countries."

Regulatory Environment

Countries with similar regulatory environments or the same regulatory bodies (like the European Union) may be grouped together as a region.

Pooled Region

These areas are typically pooled together based on certain shared characteristics, such as ethnic, genetic, environmental, or socio-economic factors, disease prevalence, or health system similarities. Thus, data from various geographies can be pooled based on the common factors or to provide a more comprehensive understanding of the drug's efficacy and safety across diverse populations.

The definition of a region is important because it determines the population that is represented in the clinical trial. It also has implications for the extrapolation of the results of the trial and their acceptance by regulatory authorities in different parts of the world. By pooling data from different regions, researchers can assess the consistency of a drug's effects and identify any potential differences in responses among various populations including a pooled region. It's a crucial part of designing and implementing an MRCT. The concept of a "pooled region" was described in the ICH E17 guideline is particularly relevant when analyzing data from MRCTs, as these trials are conducted across multiple regions with different ethnicities, cultures, and healthcare systems. However, when pooling regions, it is crucial to ensure that the grouping is justified and scientifically sound. Insufficient information and data about impact of ethnic factors to drug responses would make application of pooled concept more difficult. Factors such as the similarity of the patient populations, healthcare practices, standard of care, and regulatory requirements should be considered when deciding which regions to pool. More accumulation of data on ethnic factors through conduct of MRCTs and other related research would be useful for better interpretation of data from MRCTs [1, 2].

It is important to note, as highlighted in the previous section, various survey results have suggested that there is lack of understanding, and application, of the "pooled region".

Diversity in Clinical Trials

When the ICH E17 guideline was being drafted, the concept of "diversity in clinical trials" was not specifically highlighted or emphasized [13]. It was expected that when designing a MRCT protocol for global drug development diverse patient populations in clinical trials ensure that the study findings are applicable to people of different ages, races, ethnicities, genders, and socioeconomic backgrounds in all regions. This helps to identify any potential differences in drug response among different groups and ensure that new treatments are safe and effective for the intended population. It was also emphasized that understanding genetic and biological differences would allow to study different populations that may have varying genetic backgrounds and biological factors that influence drug response, metabolism, and safety. Including diverse participants in clinical trials allows researchers to better understand these factors and optimize drug dosing, efficacy, and safety for various populations.

In theory, MRCT could be conducted in one "region" if that region has similar influential factors that representative of the multiple regional population. In that ideal situation, the data can be used for registration in another region. This strategy was tested unsuccessfully, when the US Food and Drug Administration rejected the registration of an anticancer immunotherapy drug that was primarily developed in another single country [14]. The FDA opined that the sponsor needed to provide clinical data that is applicable to the U.S. population. This rejection by the US FDA has cited the principles of E17 as the agency and its outside advisers were critical for not enrolling a diverse enough population for the results to be applicable to U.S. patients. It should be noted, the ICH E17 guidance also recommend including various populations from different regions for examining similarities and differences of ethnic factors on drug response from an early phase of clinical trials. The FDA decision highlights the need for diversity in clinical trials and could deter drugmakers from relying on clinical tests exclusively performed in one country for a global registration.

Evolution of Decentralized Clinical Trials

The use of decentralized clinical trials (DCTs), also known as virtual or remote clinical trials, have become popular over the last few years especially during the COVID pandemic [15, 16]. Here the clinical trial relies on digital technology and innovative methods to collect data from study participants without requiring them to visit a central trial site or clinic. This approach aims to increase the accessibility, efficiency, and convenience of clinical trials for both participants and researchers. As part of remote the monitoring participants can be monitored remotely using digital devices, such as smartphones, wearable sensors, and telemedicine platforms. Telemedicine consultations and home-based interventions are also allowed.

While DCTs offer numerous benefits, they also come with potential challenges that differ across the regions, such as data privacy and security concerns, technology access disparities, and ensuring data quality and regulatory compliance. Despite these challenges, the adoption of DCTs is expected to increase as technology continues to advance and the need for more efficient and patient-centric clinical research grows. Moreover, not all regions and regulators allow for conduct of decentralized clinical trials thus creating lesser opportunities for global participating using DCT. Many of these topics related to DCT were not considered when the ICH E17 guidelines were drafted.

Increase Use of Real-World Evidence/Data (RWE/RWD)

Real-world evidence (RWE) is derived from real-world data (RWD), such as electronic health records, claims databases, patient registries, and other sources outside of controlled clinical trials. RWE has the potential to inform healthcare decision-making and complement evidence from traditional clinical trials [17, 18], although there are both advantages and disadvantages associated with RWE.

However, the ICH E17 guideline did not explicitly consider the use of RWD and RWE. For example, although data sources may vary significantly across different regions, with variations in data collection methods, data quality, and access to proprietary databases. Understanding and adhering to the specific regulations in each region can be challenging, as there may be differences in data privacy and security laws, consent requirements, and data sharing agreements. Differences in treatment guidelines, access to healthcare services, and healthcare infrastructure may introduce bias and confounding factors in the analysis of RWD. Understanding and accounting for these regional variations is critical for accurate interpretation of the findings. Finally, extrapolation of RWD/RWE across different populations and regions can be challenging.

Limitations of ICH E5 Guidelines

ICH E5 guideline Titled: Ethnic Factors in the Acceptability of Foreign Clinical Data, published in 1998 provides recommendations on how to evaluate the potential impact of ethnic factors on the safety and efficacy of drugs [3]. The ICH E5 guideline focuses on the extrapolation of foreign clinical data to a new region, but it may not be fully applicable in situations where a drug is being developed simultaneously in multiple regions or in global clinical trials. The ICH E5 guidelines are about 3 decades old, and since then, our understanding of the role of ethnic factors in drug response has evolved significantly. For example, the guideline may not fully capture the current state of knowledge in pharmacogenomics and related fields. Also, there appears to be more focus on genetics and intrinsic factors which are easily identifiable while the extrinsic factors are more difficult to study and are generally overlooked.

The definition of the region, denoted by the 'R' in MRCT, has also evolved under the E17 guidelines. Multiregional clinical trials involve conducting clinical studies in different geographic regions simultaneously. The definition of the region has expanded beyond geographical boundaries to include factors such as regulatory and ethnic considerations. The E17 guidelines provide guidance on how to interpret and apply the concept of region in the design and conduct of multiregional clinical trials, considering these broader implications.

Relating to the discussion about region, the question always remained, who is Asian? Asia is a diverse continent spanning 30% of the world's landmass and over 60% of the world's population. Thus, geographic location, race, and ethnicity by themselves are poor surrogates to determine meaningful clinical effect. A single ethnic label of "Asian" may not represent the broad diversity between collections of people in Asia. Thus, the "Asian" phenotype underestimates the genetic diversity of Asia yet potentially overstates its impact on variability in drug disposition and pharmacodynamics [19].

Not All Sponsors are Created Equal

Historically the drug development and clinical trials was "vertically integrated" whereby the sponsor, generally a multinational pharmaceutical company, would have end to end drug development in-house—starting with preclinical to post-marketing. The ICH E17 guidelines encourage to "start with end in mind" approach of designing Phase 1 studies to identify various confounding factors that would guide the subsequent phases including the pivotal studies. However, over the last two decades the drug development paradigm has changed from integrated R&D (whereby the sponsor would conduct the research leading to development) to "S&D"—Search and Develop. It is well recognized that acquisitions have played a significant role in drug development and marketing in recent decades. Pharmaceutical companies often acquire start-up or smaller biotech companies to gain access to promising drug candidates to enhance their own product portfolios. Since the smaller biotechnology companies often rely on being acquired as their exit strategy they are focus on speed with studies within same country or region rather the MRCT. This resulted in smaller companies designing Phase 1 and 2 studies with the intention of being acquired and not considering the end-to-end development approach advocated in the ICH E17 guidelines. In many cases, a small company will develop a drug to a certain stage, and then a larger company will acquire the small company and will conduct the later-stage trials and bring the drug to the market. Hence these acquired assets do not have the necessary MRCT data from various regions around the world to design meaningful pivotal MRCT studies and continue the strategy of local/regional development and registration.

Recent report from US Congressional Budget Office in 2021 show that small drug companies (those with annual revenues of less than $500 million) now account for more than 70% of the nearly 3,000 drugs in phase III clinical trials [20]. While no data are available to identify if these are being conducted using MRCT, if speed to market is the aim, then it is likely these studies are being conducted locally or regionally. These small drug companies are also responsible for a growing share of drugs already on the market. Since 2009, about one-third of the new drugs approved by the US Food and Drug Administration have been developed by pharmaceutical firms with annual revenues of less than $100 million [21]. Many of these companies have been acquired by larger companies. Large drug companies (those with annual revenues of $1 billion or more) still account for more than half of new drugs approved since 2009 and an even greater share of revenues, but they have only initiated about 20% of drugs currently in phase 3 clinical trials [22]. These larger companies have the means and expertise to conduct MRCTs, Data are lacking on how many of these large companies do conduct MRCT studies as a rule while the start-up or smaller companies do not have the resources or expertise to conduct MRCT studies with global registration strategy in mind.

Not all Products Require MRCT

Oncology, neurology, and infectious diseases have all had a rising share of new launches in the past five years with 197 of the 330 launches (60%). Precision medicine increasingly dominates in oncology where targeted therapies account for almost all of research, and over 40% of the pipeline is for rare cancers with nextgeneration biotherapeutics [23]. Thus, not all drug development may require multiregional clinical trials. For example, cell and gene therapies are generally not amenable to MRCT rather are more suitable for small multi-center or even as part of cancer network such as NCCN (National Comprehensive Cancer Network). Previous studies have shown that usually large outcome studies for CV or diabetes have effectively utilized the end-to-end MRCT [24].

Recommendations on How to Improve the Uptake of ICH E17 Guidelines

E17 Pooling Strategy and Its Application

One of the key concepts introduced in the ICH E17 is the pre-specified pooling strategy (see Principle #4 in the ICH E17 [1]). This Principle aims to provide flexibility in sample size allocation to regions, facilitate the assessment of consistency in treatment effects across regions, and support regulatory decision-making. The pooling strategy should be based on established knowledge about key intrinsic/extrinsic factor and effect modifiers (i.e., those intrinsic/extrinsic factors that impact the treatment effects). The identification of these factors should be collected and examined early in the drug development program, and further be reflected in the study design at late stage (ICH E17 Training Module 2) [2].

Based on these systematic approaches of collecting, examination, to reflection on intrinsic/extrinsic factors, study subjects who have similar factors shall be defined prospectively in late phase trials, either as pooled region (e.g., pooling patients from East Asia Countries with similar confounding factors) or pooled subpopulation (e.g., pooling patients with particular polymorphism). The consistency evaluation should be structured and can take a layered approach that examine treatment effects at country level, region level, subpopulation level and global level. To enable the holistic evaluation, the sample size allocation can focus on the pooled region and/or pooled subpopulation, balanced with the considerations of country level regulatory need [25] (Also, refer to ICH E17 training modules 4 to 6 [2]).

It should be specially noted that early phase studies, such PK/PD studies or epidemiology studies, can provide valuable information to define the pooling regions/subpopulations. The 2014 guideline “Basic principles for conducting phase 1 trials in the Japanese population on prior to global clinical trials” in Japan has given specific consideration for conducting early phase trials in the region [26]. The NMPA Center for Drug Evaluation (CDE) is developing related guidelines, including considerations for conducting early phase trials in China prior to confirmatory MRCT. The strategies for conducting a phase I trial in China should be comprehensively evaluated during the clinical development plan. This guideline can facilitate guiding the related work and promote China to participate in global development as soon as possible [25]. A future ICH E17 Q&A document may be useful to provide a harmonized guidelines on the connection between early and late phase trials for MRCT design and interpretation of the current guidelines.

Apply Lessons from the Pandemic on Conduct of MRCT and ICH E17 Guidelines

The vaccines developed during the pandemic applied many of the principles outlined in the ICH E17 guidelines and there are many lessons learned from the use of MRCTs which should be made business as usual [27]. For example, the pandemic underscored the need for greater harmonization of regulatory requirements across countries. Differences in regulatory requirements did not slow down the initiation and progression of MRCTs. The pandemic also accelerated the adoption of remote monitoring and other digital health technologies in MRCTs. These technologies can enhance the efficiency of MRCTs even post-pandemic. The changing conditions of a pandemic require flexibility in trial design. Adaptive trial designs, which allow modifications to the trial after it begins (based on interim results), have proved to be particularly beneficial. As the pandemic evolved, participation in MRCTs provided the flexibility to pivot resources and focus to regions where the pandemic's impact was greater. Frequent communication with the regulators as recommended in the E17 guidelines provided guidance to sponsors on how to manage clinical trials during the pandemic, addressing issues such as protocol deviations, remote monitoring, and patient safety. Despite the need for speed the regulators ensured that the study findings were applicable to populations of different ages, races, ethnicities, genders, and socioeconomic backgrounds for drug approval. This helped to identify any potential differences in drug response among different groups and reduce disparities in healthcare access and outcomes.

While not discussed in the ICH E17 guideline, the pandemic provided examples of Emergency Use Authorization (EUA) for the use of medical products that have not undergone the usual rigorous testing and approval process. Likewise, there were Conditional Approval where a medicinal product received conditional approval based on less complete clinical data and iterative phases of data gathering than normally required if the drug fills an unmet medical need for a serious condition and had positive impact on public health. All these pathways accelerated the drug development and approval process which can be weaved in the future design and conduct of MRCT. However, as noted earlier, the ICH E17 guidelines were written whereby intrinsic and extrinsic data would be generated in earlier phases of the clinical development to help with design of pivotal studies. Due to urgency and speed of drug development during the pandemic, many of the intrinsic and extrinsic factors were not collected in early phase of development During the pandemic the sponsors and the regulators were not necessarily thinking "how do I apply ICH E17 guidelines" rather their focus to how to simultaneously develop vaccines and life-saving therapies around the world. The focus was on speed, quality, and consistency-all the factors advocated in the ICH E17 guidelines.

Adaptive Clinical Trial Design and Its Application to MRCTs

Adaptive clinical trial design has become increasingly popular to increase productivity of drug development. This is further fueled by the urgency brought by the pandemic. New ICH guidelines (ICH E20) are being developed on the design, conduct, analysis, and interpretation of adaptive clinical trials. Even though ICH E17 did not specifically discuss this topic, more and more MRCTs are using adaptive design, especially group sequential design for global simultaneous drug/vaccine development and registration. A few additional points may need to be considered: (1) Aligning with the ICH E17 principles, the interim analyses that have the potential to stop the trial early for multi-regional filings may need to examine the regional consistency as part of the key supportive analyses; (2) the data monitoring committee (DMC) may need to have representations, not only from different functional expertise (e.g., clinical, statistics), but also from key regions; (3) In case the trial results are considered insufficient (e.g., limited sample size/exposure or total amount of statistical information such as event counts) for some regions, and if the MRCT has a regional extension, the proper firewall should be in place to maintain the trial integrity.

Focus on Social Determinants of Health for Extrinsic Factors

Historically there has been too much focus on ICH E5 guidelines with the regulators and sponsors about impacts of ethnic factors, especially intrinsic factors such as genetics since the genetic-factors are often more quantifiable and easier to study than many non-genetic factors. However, as our understanding of drug response continues to evolve, the field is increasingly recognizing the importance of considering a broad range of Social Determinants of Health [28], not just genetic ones, to truly optimize drug therapy for different populations. Drug development is a multifactorial complex process and the ICH E5 guidelines should be taken into context of new definition of 'region' in MRCT as defined in the E17 guidelines that encourage pooling of data across regions especially focusing on the determinants of health.

Increase Focus on Patient Reported Outcomes

Patient Reported Outcomes (PROs) offer insights into how a treatment is experienced by patients across various regions, providing valuable data on the effect of cultural, geographical, or socioeconomic factors on patients' perceptions of their health and treatment. PRO measures would need to be standardized across multiple regions to provide consistent data, even though the trials are conducted in diverse geographic and cultural settings and would provide holistic understanding of treatment impact. Finally, the PROs can help regulatory authorities make more informed decisions about the approval and labeling of drugs and treatments [29–31]. This data would provide a more complete picture of the benefits and risks of a treatment from the patients' perspective. This would require consistent data collection across multiple regions, each with its unique language. The translation and cultural adaptation of PRO measures need to be done carefully to maintain their validity. It is also critical to ensure that data privacy regulations are adhered to when collecting and processing PRO data from different regions. There is no mention of this in the current ICH E17 guidelines but can be considered in design of the MRCTs.

Pivot Towards Digital

The pivot towards digital technologies has been a significant development of MRCTs, particularly during the pandemic. As travel and face-to-face contact were heavily restricted due to safety concerns, digital technologies proved invaluable in facilitating continued clinical research. This has manifested itself in enhanced remote monitoring and data collection using electronic informed consent (eIC) to reduce the need for physical paperwork and in-person meetings. There was increase in telemedicine through virtual visits with trial participants, assessing their condition, providing advice, and even delivering some forms of treatment remotely. If telemedicine is to become a part of MRCTs, new methodology would need to be created to ensure the data quality with stricter QA/QC checks. Many MRCTs are moving towards a decentralized model, where many aspects of the trial, such as patient recruitment, data collection, and even some aspects of treatment, are conducted remotely using digital tools. There is increased use of artificial intelligence (AI) and machine learning (ML) technologies for data analysis, pattern recognition, and predictive modeling, making it possible to quickly analyze the vast amounts of data generated by MRCTs and gain insights more quickly. Finally, the cloud-based data storage and sharing digital storage solutions will make it easier to store and share the large volumes of data generated by MRCTs, improving collaboration, and enabling more rapid analysis. While the pivot towards digital technologies had many benefits, it also brings new challenges. Issues such as data security, privacy, and the digital divide (where some populations may not have access to the required technology) need to be carefully managed to ensure that MRCTs are conducted ethically and effectively. Incorporating this in consistent manner across the various regions would allow to effectively incorporate digital methodologies into MRCT design and conduct.

Update the ICH E17 Guidelines Through Q&A

One of the ways ICH maintains and improves upon its existing guidelines is through a Questions & Answers (Q&A) process. This process allows for clarification, interpretation, and expansion of the guidance provided in the guidelines based on questions or issues raised by stakeholders. Historically the Q&A are developed five to seven years after the original publication. With the Q&A the ICH guidelines can remain current and relevant to be interpreted and applied consistently across different regions. It also would allow for a level of flexibility and adaptation in response to emerging issues or advances in scientific understanding.

Conclusion

It has been over 5 years since the ICH E17 guidelines on MRCT were adopted and training material produced by ICH. Much has changed on how drugs are developed and registered, especially some novel approaches were adopted during the pandemic which were not originally considered when the ICH E17 guideline was developed. In response to these changes, there are still topics that require clarification, such as pooling strategies, local vs. regional data for registration, and other operational challenges.

In addition, the pandemic has shown us that newer approaches adopted by sponsors and regulators can be used to perform clinical trials and approve medicines. This has contributed to shorter registration timelines and more streamline clinical studies. The question now remains can we weave-in the principles of ICH E17 MRCT into the drug development and registration approach for the next decade. This may be a good time to start the ICH Q&A process to provide more contemporary guidance.

Author Contributions

All authors (except JW) were members of the expert working group for ICH E17 guideline published in 2017. The views expressed in this article are those of the authors and do not necessarily reflect the official views of the Pharmaceuticals and Medical Devices Agency, National Medical Products Administration or those of companies that employ them.

Funding

No funding was received for this work.

Declarations

Conflict of interest

RS, WW, AC work for for-profit biopharmaceutical companies. The work by AC was performed as the Senior Statistical Advisor, Office of the Commissioner, US Food and Drug Administration. JW and YU are regulators who work for their respective regulatory agencies.

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